Lecture 20: Dual Supergradient Ascent

MA3236 Non-Linear Programming

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Quick review: from a saddle point to a dual algorithm

$$\max_{\mathbf{w}\in W} \theta(\mathbf{w}), \qquad \theta(\mathbf{w}) = \inf_{\mathbf{x}\in\mathbf{R}^n} L(\mathbf{x},\mathbf{w}) = \inf_{\mathbf{x}\in\mathbf{R}^n} \{f(\mathbf{x}) + \mathbf{w}^T \boldsymbol{r}(\mathbf{x})\},$$

$$\mathbf{w} := \begin{pmatrix} \boldsymbol{\lambda} \\ \boldsymbol{\nu} \end{pmatrix}, \quad W := \mathbf{R}^m \times \mathbf{R}^p_+, \quad \boldsymbol{r}(\mathbf{x}) := \begin{pmatrix} \mathbf{g}(\mathbf{x}) \\ \mathbf{h}(\mathbf{x}) \end{pmatrix} \in \mathbf{R}^{m+p}.$$

- L18: θ is concave; a finite dual value is a primal lower bound.
- L19: strong duality can make that bound exact. A saddle point alone does not make naive descent–ascent converge.

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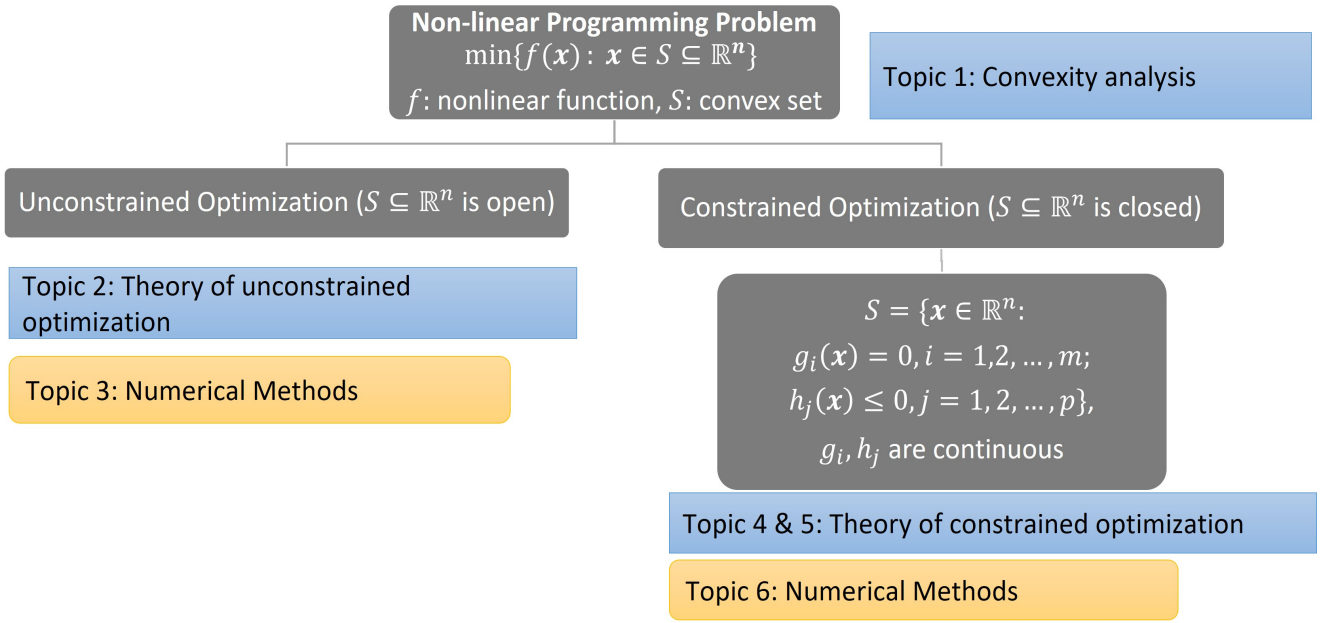
- L18: θ is concave; a finite dual value is a primal lower bound.
- L19: strong duality can make that bound exact. A saddle point alone does not make naive descent–ascent converge.

This lecture:

$$\mathbf{w}^{(k)} \longrightarrow \mathbf{x}^{(k)} \in \arg\min_{\mathbf{x}} L(\mathbf{x}, \mathbf{w}^{(k)}) \longrightarrow \mathbf{w}^{(k+1)}.$$

The main difficulty: how to define a "gradient"?

Topics



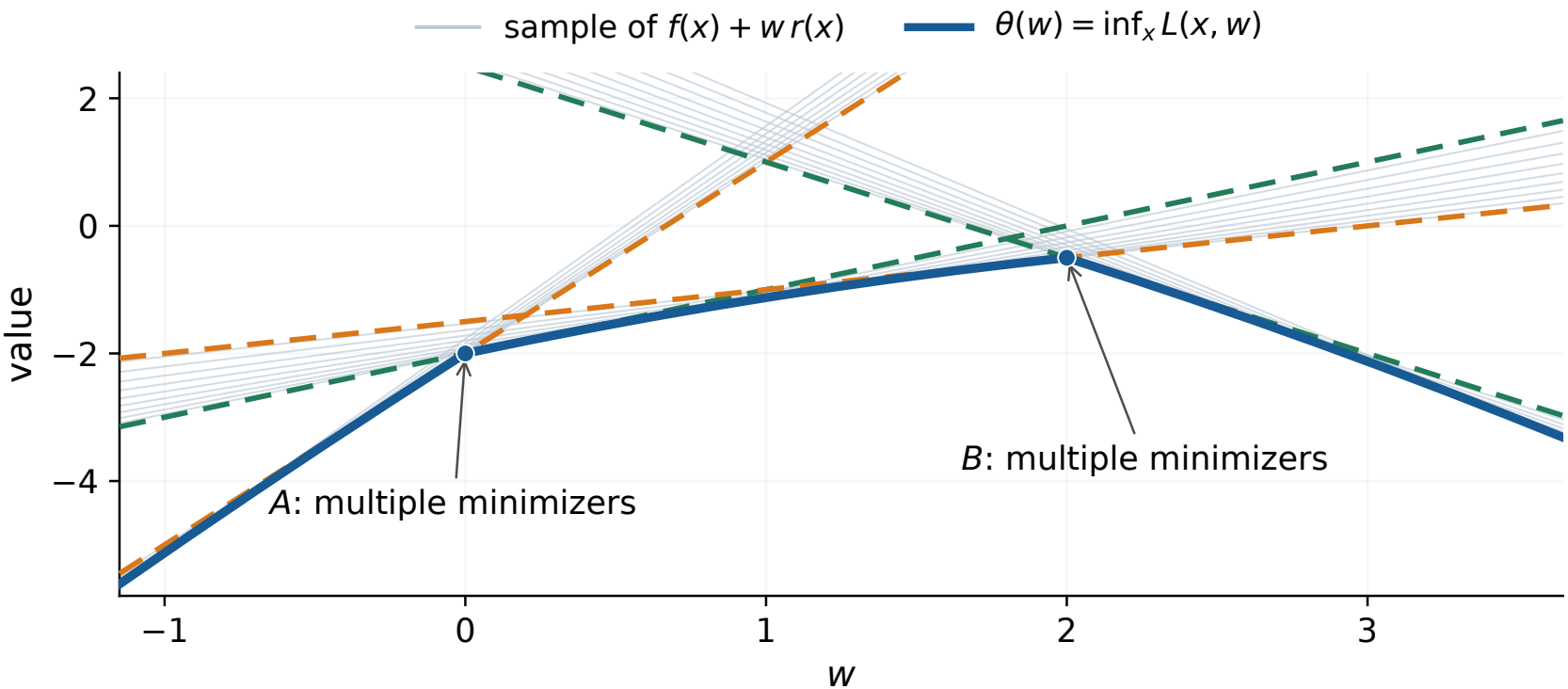
In Topic 6, we study numerical methods for solving **constrained optimization problems**.

- Quadratic penalty and augmented Lagrangian methods
- Lagrange–Newton and sequential quadratic programming
- Barrier methods
- Dual supergradient methods

An infimum can create a corner

For fixed $\mathbf{x}$, $L(\mathbf{x}, \mathbf{w}) = f(\mathbf{x}) + \mathbf{w}^T \boldsymbol{r}(\mathbf{x})$ is affine in $\mathbf{w}$.

$$\theta(\mathbf{w}) = \inf_{\mathbf{x} \in \mathbf{R}^n} \{f(\mathbf{x}) + \mathbf{w}^T \boldsymbol{r}(\mathbf{x})\}.$$

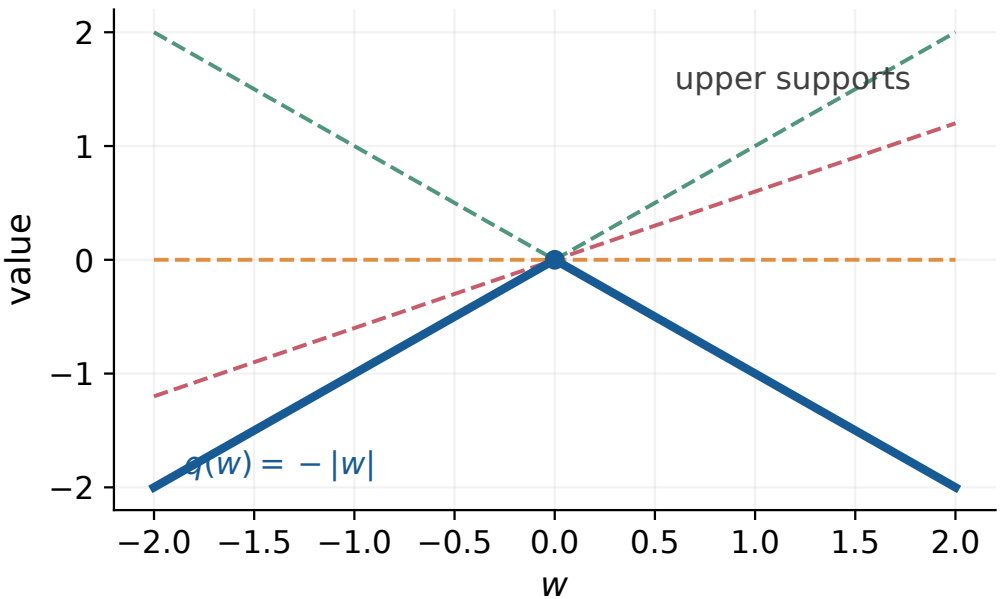


Supergradients: supporting planes from above

For concave q, at a point where $q(\mathbf{w})$ is finite, a supergradient satisfies

$$q(\mathbf{z}) \le q(\mathbf{w}) + \boldsymbol{\xi}^T(\mathbf{z} - \mathbf{w}) \quad \text{for every } \mathbf{z}.$$

The set of all such vectors is the superdifferential $\partial^+ q(\mathbf{w})$.



$q(w) = -|w|$:

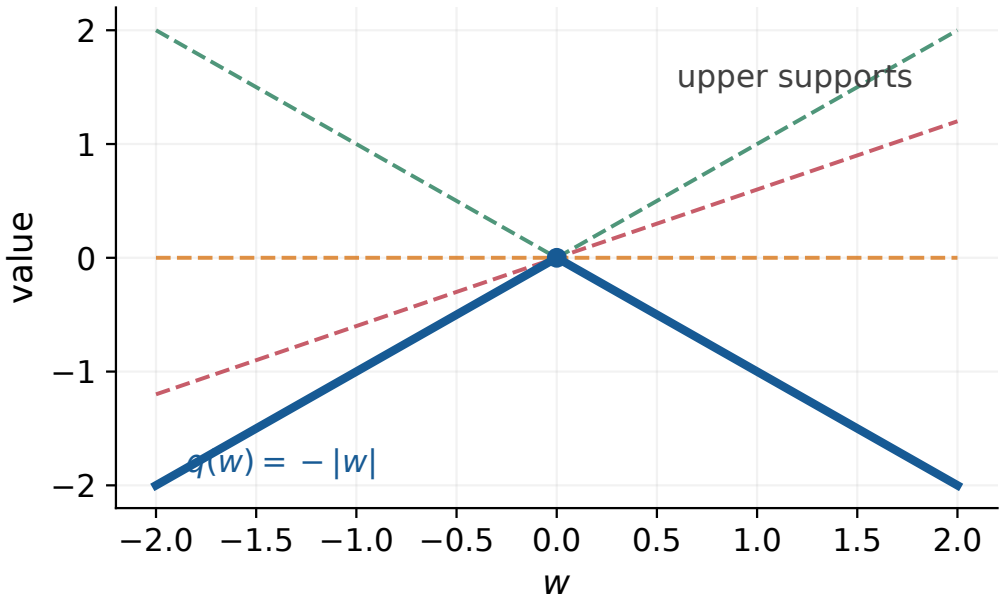
$$\partial^+ q(w) = \begin{cases} \{1\}, & w < 0, \\ [-1, 1], & w = 0, \\ \{-1\}, & w > 0. \end{cases}$$

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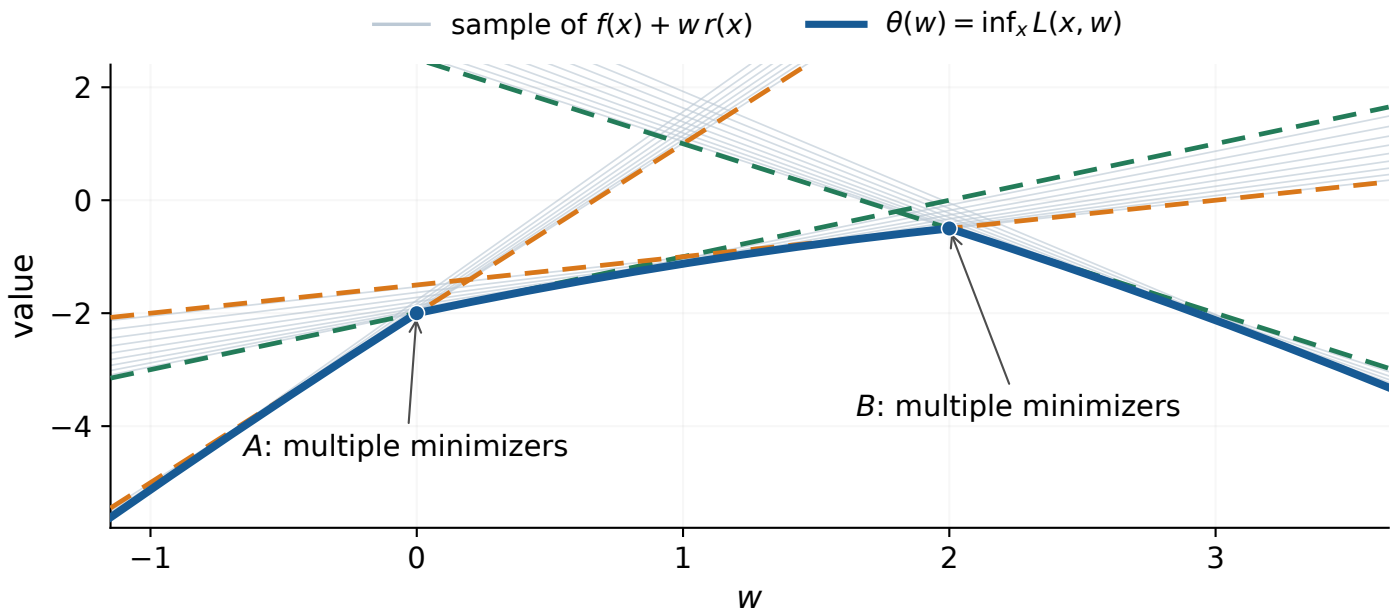
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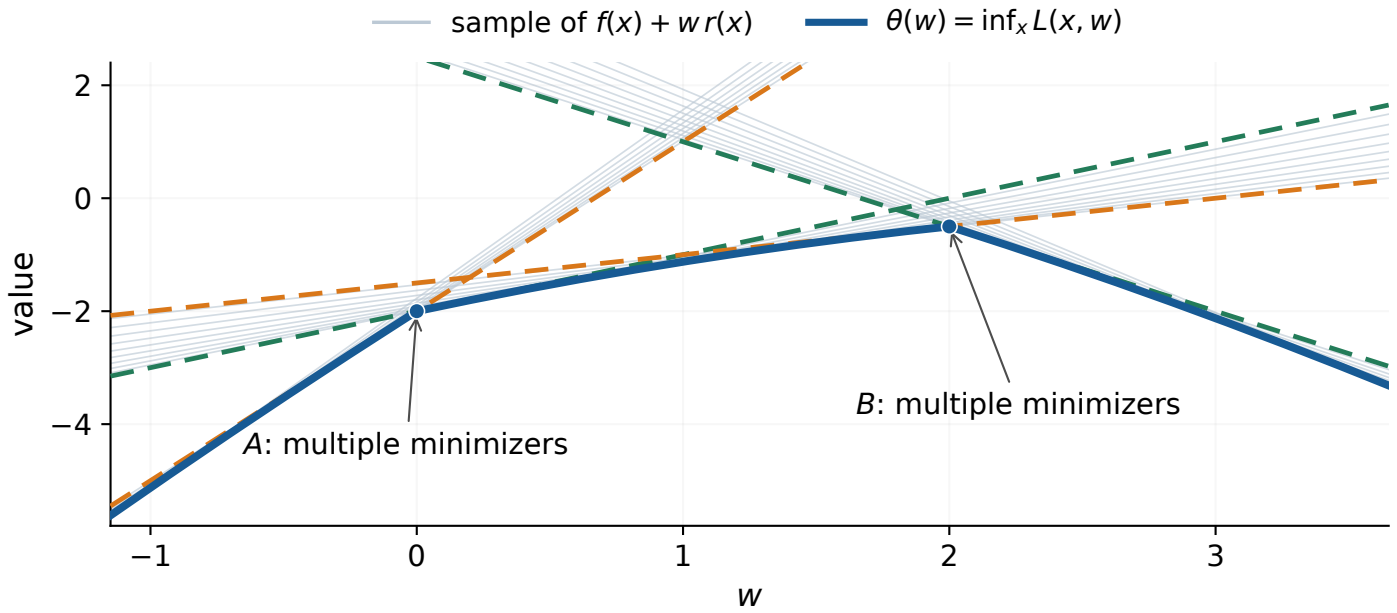
For concave q finite near $\mathbf{w}$:

q differentiable at $\mathbf{w} \iff \partial^+ q(\mathbf{w})$ is a singleton.

An inner solution supplies a supergradient



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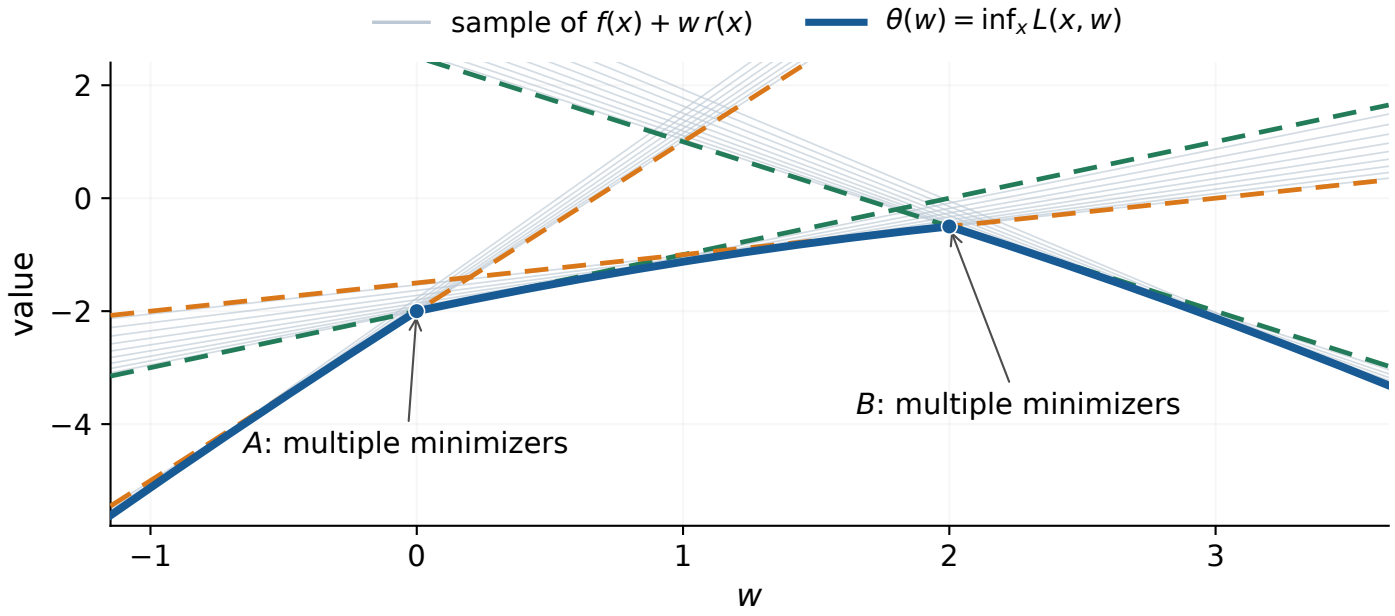
Theorem 20.1 (Residual supergradient)

Write $X(\mathbf{w}) = \arg\min_{\mathbf{x}\in\mathbf{R}^n} L(\mathbf{x}, \mathbf{w})$. *For any* $\mathbf{x} \in X(\mathbf{w})$,

$$\boldsymbol{r}(\mathbf{x}) \in \partial^+\theta(\mathbf{w}).$$

NB. One global inner minimizer is enough; uniqueness is unnecessary.

Some properties of the superdifferential



Theorem 20.2 (Danskin's formula)

Suppose $f, \boldsymbol{r}$ are continuous. In an open neighborhood of $\mathbf{w}$, all inner minima are attained and all their minimizers lie in one compact set. Then

$$\partial^+\theta(\mathbf{w}) = \operatorname{conv}\{\boldsymbol{r}(\mathbf{x}) : \mathbf{x} \in X(\mathbf{w})\}.$$

Here "conv" means all convex combinations, or in short, convex hull.

Some properties of the superdifferential (cont'd)

Danskin's formula: under the preceding assumptions,

$$\partial^+\theta(\mathbf{w}) = \text{conv}\{\boldsymbol{r}(\mathbf{x}) : \mathbf{x} \in X(\mathbf{w})\}.$$

Corollary of Danskin's formula: iff for differentiability.

θ differentiable at $\mathbf{w}$ $\iff$ all $\mathbf{x} \in X(\mathbf{w})$ have the same residual.

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Corollary of Danskin's formula: iff for differentiability.

$$\theta \text{ differentiable at } \mathbf{w} \iff \text{all } \mathbf{x} \in X(\mathbf{w}) \text{ have the same residual.}$$

Sufficient condition for differentiability: $X(\mathbf{w}) = \{\mathbf{x}(\mathbf{w})\}$ is a singleton.

Example 20.3 (Several minimizers, one residual)

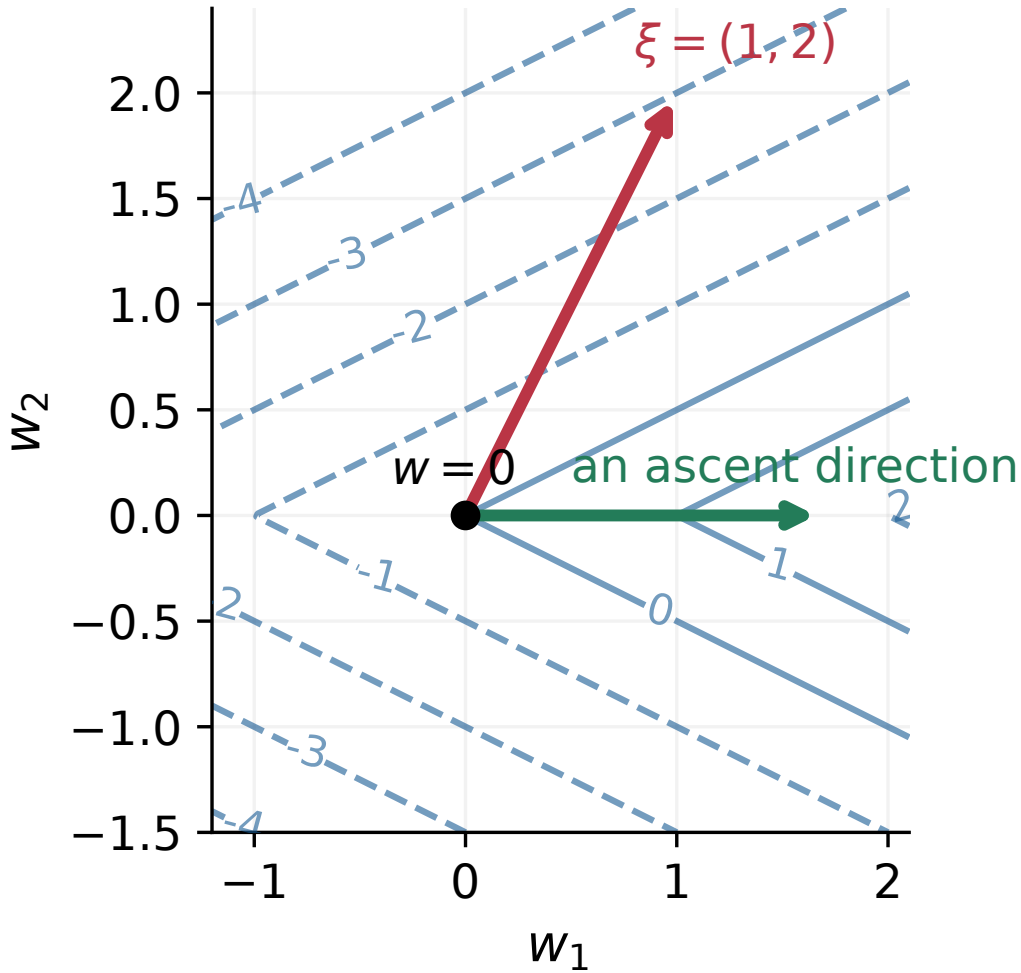
$$L(\mathbf{x}, \lambda) = \tfrac{1}{2}x_1^2 + (x_2^2 - 1)^2 + \lambda x_1 = \tfrac{1}{2}(x_1 + \lambda)^2 + (x_2^2 - 1)^2 - \tfrac{1}{2}\lambda^2.$$

$$X(\lambda) = \{(-\lambda, 1), (-\lambda, -1)\}, \qquad \theta(\lambda) = -\tfrac{1}{2}\lambda^2.$$

Both minimizers have residual $g(\mathbf{x}) = x_1 = -\lambda$.

A supergradient need not point uphill

Consider $q(w_1, w_2) = w_1 - 2|w_2|$ at $\mathbf{w} = \mathbf{0}$.



$$\partial^+ q(\mathbf{0}) = \{(1, t) : |t| \leq 2\}.$$

Choose $\boldsymbol{\xi} = (1, 2)$. Then

$$q(\alpha\boldsymbol{\xi}) = -3\alpha < 0 \quad (\alpha > 0).$$

Every positive step goes downhill, even though this point is not optimal.

Which directions really increase the function?

For a concave function finite near $\mathbf{w}$,

$$q'(\mathbf{w};\mathbf{d}) = \lim_{t \downarrow 0} \frac{q(\mathbf{w}+t\mathbf{d}) - q(\mathbf{w})}{t}$$

Concavity $\Rightarrow q'(\mathbf{w};\mathbf{d}) \in \mathbf{R}$.

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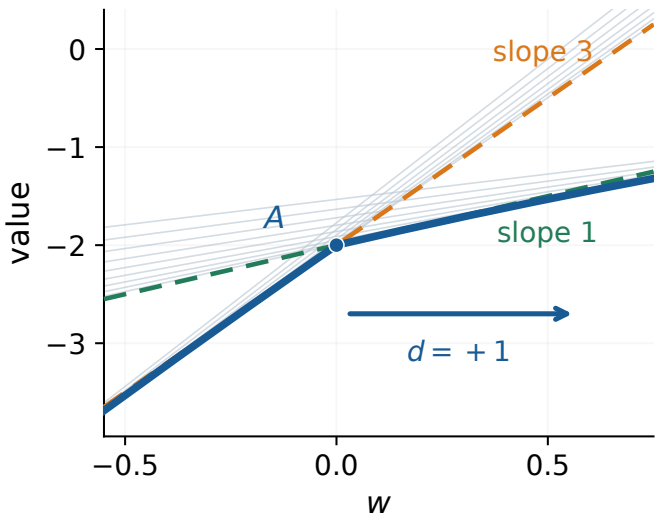
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Concavity $\Rightarrow q'(\mathbf{w};\mathbf{d}) \in \mathbf{R}$. The worst supporting slope matters!

Example 20.4

Consider $q(w) = -|w|$. Then $q'(0;1) = q'(0;-1) = -1$.



Which directions really increase the function?

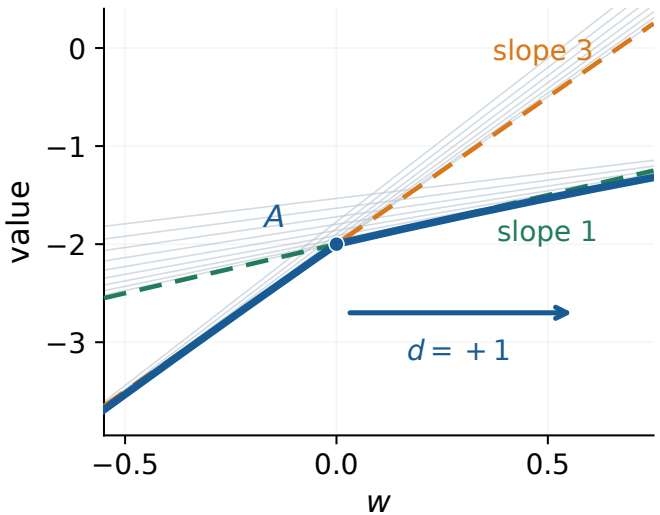
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Example 20.4

Consider $q(w) = -|w|$. Then $q'(0;1) = q'(0;-1) = -1$.



An ascent direction asks for $q'(\mathbf{w};\mathbf{d}) > 0$

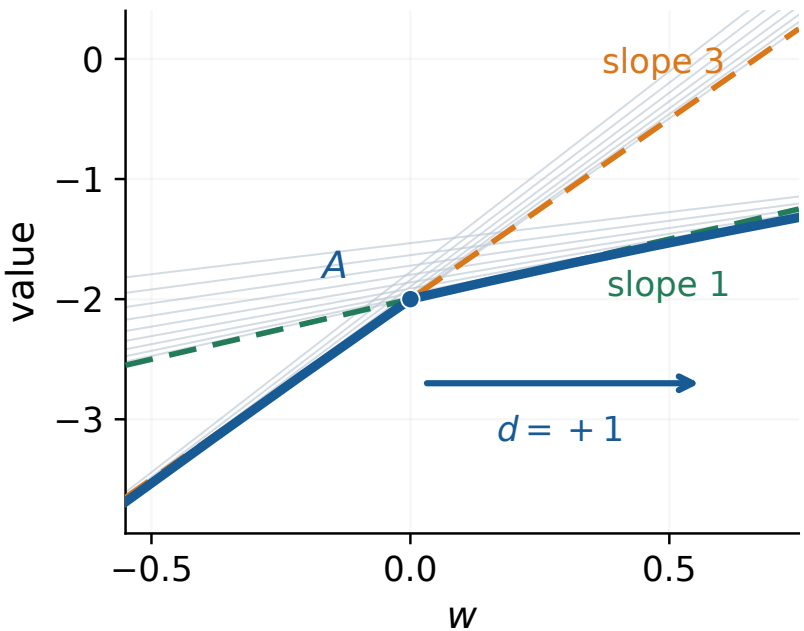
$$\Longrightarrow \quad \boldsymbol{\xi}^T \mathbf{d} > 0, \quad \forall\, \boldsymbol{\xi} \in \partial^+ q(\mathbf{w}).$$

Does it exist?

The steepest ascent direction

For concave q finite near $\mathbf{w}$, maximize $q'(\mathbf{w}; \mathbf{d})$ over $\|\mathbf{d}\| \leq 1$.

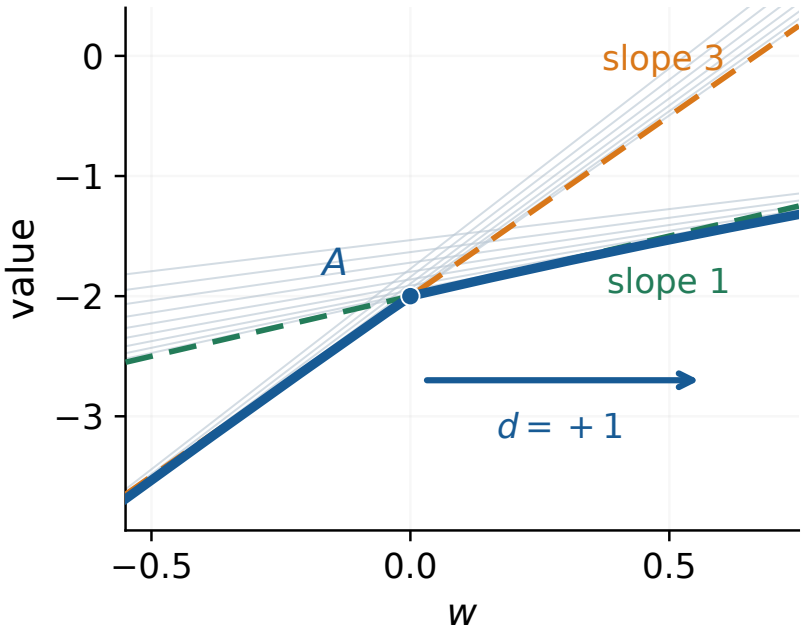
$$\max_{\mathbf{d}:\|\mathbf{d}\|\leq 1} \min_{\boldsymbol{\xi}\in\partial^+ q(\mathbf{w})} \boldsymbol{\xi}^T \mathbf{d}.$$



The steepest ascent direction

For concave q finite near $\mathbf{w}$, maximize $q'(\mathbf{w};\mathbf{d})$ over $\|\mathbf{d}\| \le 1$.

$$\max_{\mathbf{d}:\|\mathbf{d}\|\le 1} \min_{\boldsymbol{\xi}\in\partial^+ q(\mathbf{w})} \boldsymbol{\xi}^T \mathbf{d}.$$



One maximizing direction is

$$\mathbf{d}_{\text{steep}} = \begin{cases} \widehat{\boldsymbol{\xi}}/\|\widehat{\boldsymbol{\xi}}\|, & \widehat{\boldsymbol{\xi}} \neq \mathbf{0}, \\ \mathbf{0}, & \widehat{\boldsymbol{\xi}} = \mathbf{0}, \end{cases}$$

where

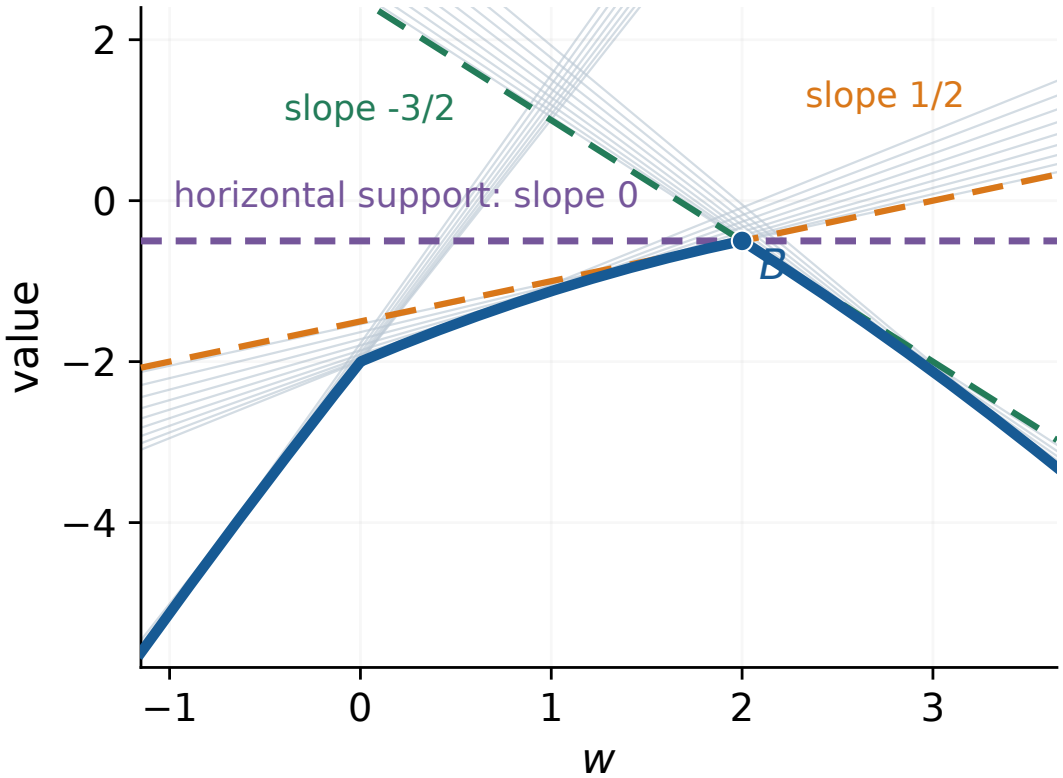
$$\widehat{\boldsymbol{\xi}} = \operatorname*{arg\,min}_{\boldsymbol{\xi}\in\partial^+ q(\mathbf{w})} \|\boldsymbol{\xi}\|.$$

The maximal slope is $\|\widehat{\boldsymbol{\xi}}\|$.

A zero supergradient gives a global upper bound

Theorem 20.5 (Nonsmooth unconstrained optimality)

$$\mathbf{0} \in \partial^+ q(\mathbf{w}^*) \iff q(\mathbf{w}^*) \geq q(\mathbf{w}) \quad \textit{for every } \mathbf{w}.$$



$$\partial^+\theta(2) = [-3/2, 1/2].$$

Since $0 \in \partial^+\theta(2)$, there is a horizontal supporting line above the entire curve.

$$\theta(w) \leq \theta(2) \quad \forall w.$$

Thus B is a global maximum.

What does the basic algorithm choose?

Idealized steepest ascent	Basic supergradient method
Uses the full $\partial^+\theta(\mathbf{w})$	
Solves $\min_{\boldsymbol{\xi}\in\partial^+\theta(\mathbf{w})} \|\boldsymbol{\xi}\|$	
$\partial^+\theta(\mathbf{w}) = \text{conv}\{\boldsymbol{r}(\mathbf{x}) : \mathbf{x} \in X(\mathbf{w})\}$	

- Practice: describing the full superdifferential and finding its least-norm point can be expensive.

What does the basic algorithm choose?

Idealized steepest ascent	Basic supergradient method
Uses the full $\partial^+\theta(\mathbf{w})$	Needs only $\mathbf{x}(\mathbf{w}) \in X(\mathbf{w})$
Solves $\min_{\boldsymbol{\xi}\in\partial^+\theta(\mathbf{w})} \|\boldsymbol{\xi}\|$	
$\partial^+\theta(\mathbf{w}) = \text{conv}\{\boldsymbol{r}(\mathbf{x}) : \mathbf{x} \in X(\mathbf{w})\}$	Simply evaluates $\boldsymbol{r}(\mathbf{x})$

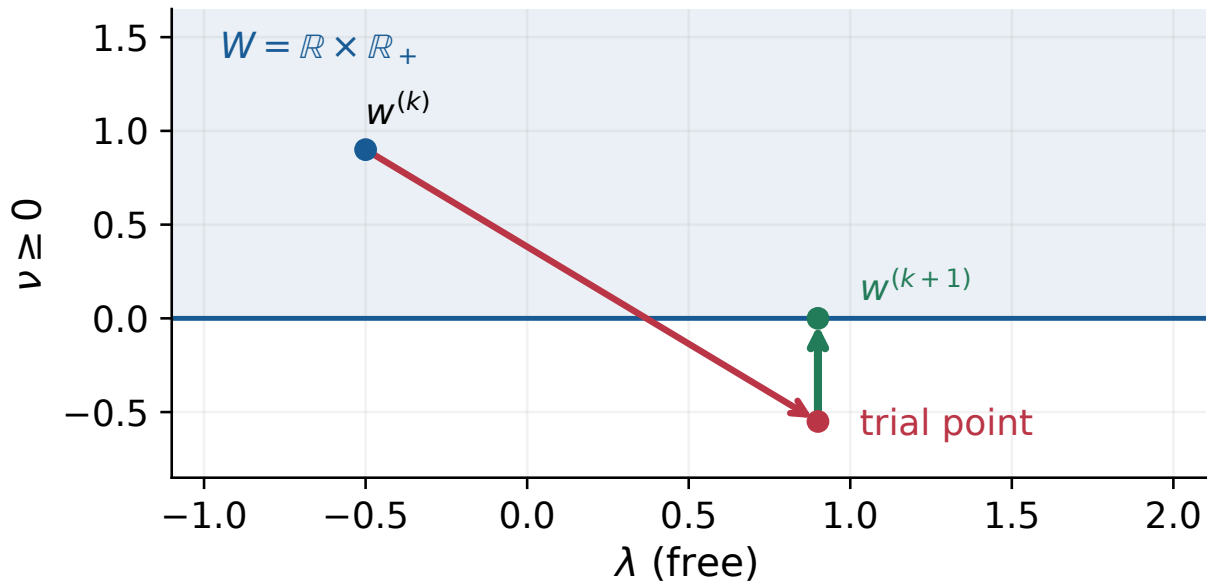
- Practice: describing the full superdifferential and finding its least-norm point can be expensive.
- Theory: bounded supergradients + nonsummable diminishing steps.

Projected dual supergradient ascent

Assume $\theta(\mathbf{w}) > -\infty$ and $X(\mathbf{w}) \neq \varnothing$ throughout W.

$$\mathbf{w}^{(k+1)} = \Pi_W\left(\mathbf{w}^{(k)} + \alpha^{(k)} \boldsymbol{r}(\mathbf{x}^{(k)})\right), \qquad \mathbf{x}^{(k)} \in X(\mathbf{w}^{(k)}).$$

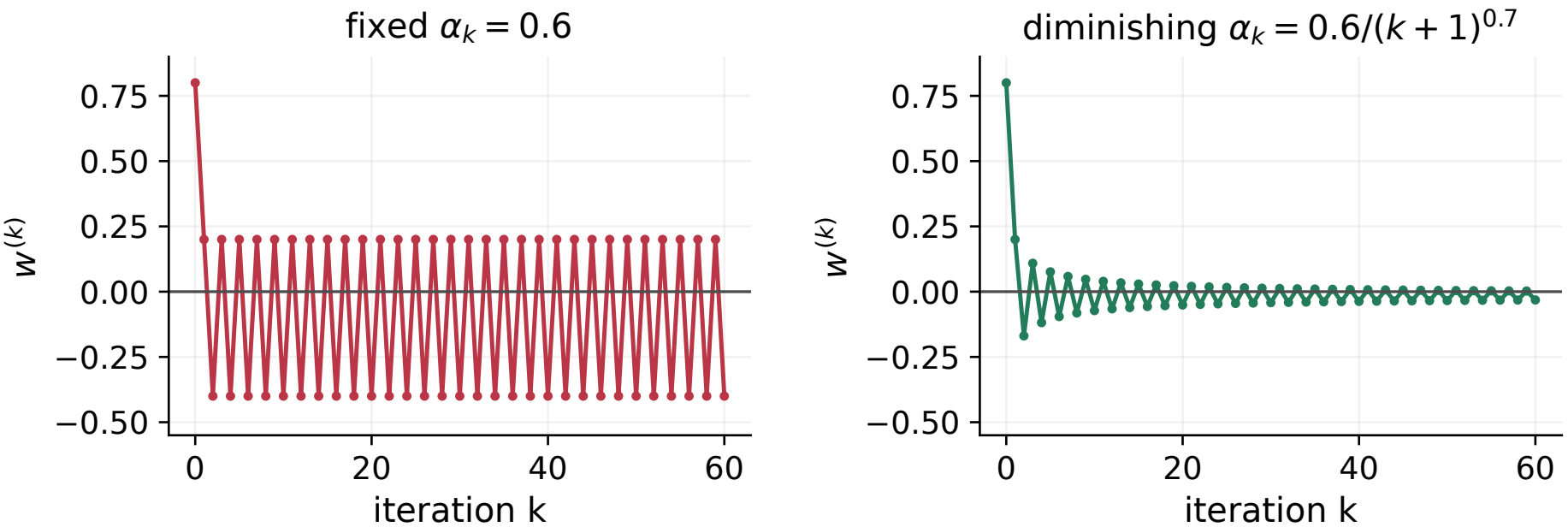
$$\Leftrightarrow \begin{cases} \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha^{(k)} \mathbf{g}(\mathbf{x}^{(k)}), \\ \boldsymbol{\nu}^{(k+1)} = [\boldsymbol{\nu}^{(k)} + \alpha^{(k)} \mathbf{h}(\mathbf{x}^{(k)})]_+. \end{cases}$$



Python: `120_dual_subgradient.py`.

Why a fixed step can fail

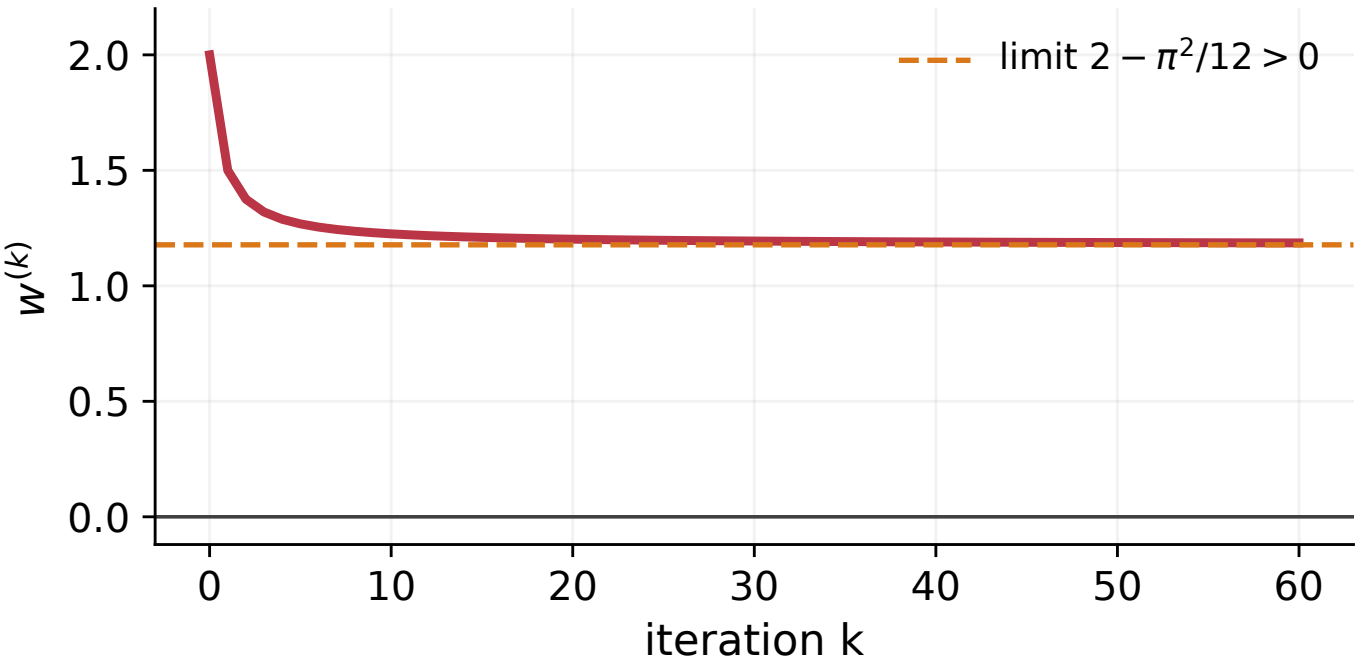
Maximize $q(w) = -|w|$: use $\xi = -\operatorname{sign}(w)$ away from zero.



- Fixed steps can overshoot the corner forever.
- Diminishing steps let those overshoots shrink.

Diminishing too quickly can also fail

Again maximize $-|w|$, now with $w^{(0)} = 2$ and $\alpha^{(k)} = \frac{1}{2(k+1)^2}$.



The total travel budget is finite and too small to reach zero.

Convergence theory

Assume a dual optimizer $\mathbf{w}^*$ exists, steps $\alpha^{(k)} > 0$, exact supergradients satisfy $\|\boldsymbol{\xi}^{(k)}\| \le G$, and $\|\mathbf{w}^{(0)} - \mathbf{w}^*\| \le R$. The projection set is closed, convex, contains $\mathbf{w}^*$, and θ is finite on it.

Theorem 20.6 (Dual objective guarantee)

For either $v_K = \theta_{\text{best}}^{(K)}$ or $v_K = \theta(\bar{\mathbf{w}}^{(K)})$, where

$$S_K = \sum_{k=0}^{K-1} \alpha^{(k)}, \quad \theta_{\text{best}}^{(K)} = \max_{0 \le k < K} \theta(\mathbf{w}^{(k)}), \quad \bar{\mathbf{w}}^{(K)} = \frac{1}{S_K} \sum_{k=0}^{K-1} \alpha^{(k)} \mathbf{w}^{(k)},$$

it holds that

$$0 \le d^* - v_K \le B_K := \frac{R^2 + G^2 \sum_{k=0}^{K-1} (\alpha^{(k)})^2}{2 S_K}.$$

Sufficient step rules

$$S_K \to \infty, \qquad \frac{\sum_{k=0}^{K-1} (\alpha^{(k)})^2}{S_K} \to 0 \quad \Longrightarrow \quad B_K \to 0.$$

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We can choose

$$\alpha^{(k)} = \frac{a}{(k+1)^q}, \qquad a > 0, \quad q \in [\tfrac{1}{2}, 1].$$

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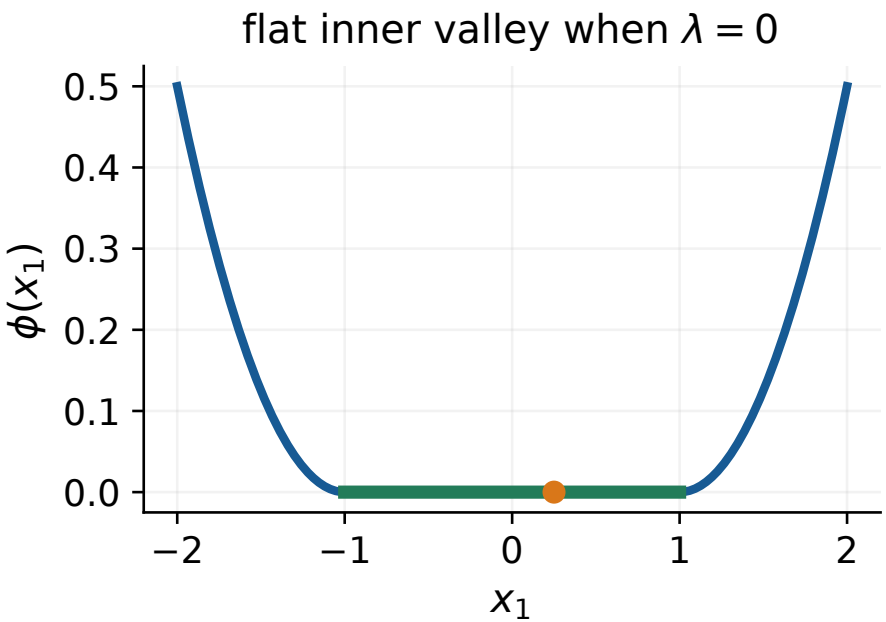
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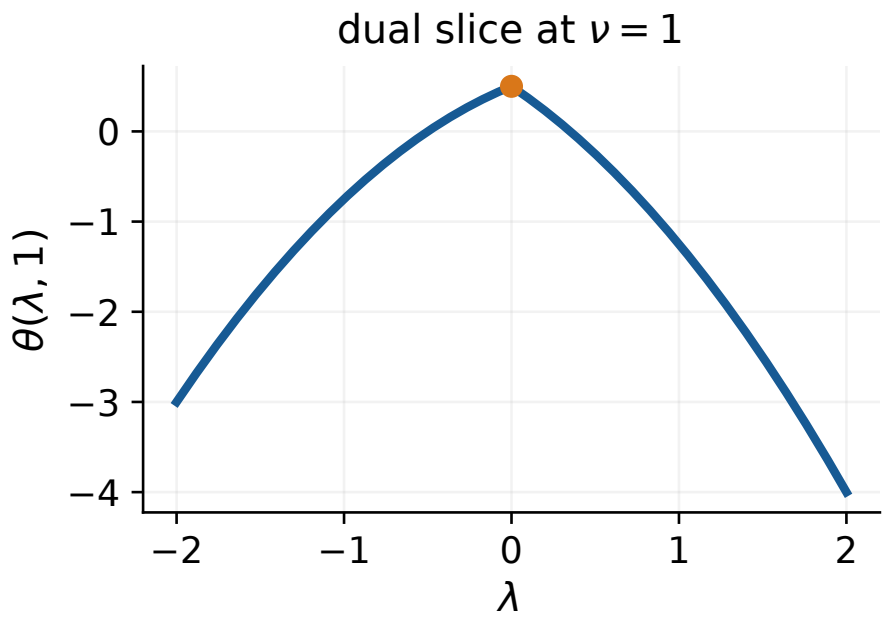
Practice:

- K-dependent constant step: $\alpha = R/(G\sqrt{K})$ gives $B_K = RG/\sqrt{K}$ for $R, G > 0$.
- Square-summable, nonsummable steps also support stochastic convergence under suitable noise assumptions.

A common test: smooth primal, nonsmooth dual

$$\min_{\mathbf{x}\in\mathbf{R}^2} \ \tfrac{1}{2}\big(\max\{|x_1|-1,0\}\big)^2 + \tfrac{1}{2}x_2^2 \quad \text{s. t.} \quad x_1 - \tfrac{1}{4} = 0, \quad 1 - x_2 \le 0.$$





The objective is convex and continuously differentiable.

$$\mathbf{x}^* = (\tfrac{1}{4}, 1), \quad \mathbf{w}^* = (0, 1), \quad p^* = d^* = \tfrac{1}{2}.$$

The exact oracle in the common test

$$L(\mathbf{x}, \lambda, \nu) = \tfrac{1}{2}\big(\max\{|x_1| - 1, 0\}\big)^2 + \tfrac{1}{2}x_2^2 + \lambda(x_1 - \tfrac{1}{4}) + \nu(1 - x_2).$$

One global inner minimizer has $x_2(\nu) = \nu$ and

$$x_1(\lambda) = \begin{cases} -\operatorname{sign}(\lambda) - \lambda, & \lambda \neq 0, \\ -1, & \lambda = 0 \text{ (fixed choice)}, \end{cases}$$

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$$\Longrightarrow \quad \theta(\lambda, \nu) = L(\mathbf{x}(\lambda, \nu), \lambda, \nu) = -|\lambda| - \tfrac{1}{2}\lambda^2 - \tfrac{1}{4}\lambda + \nu - \tfrac{1}{2}\nu^2.$$

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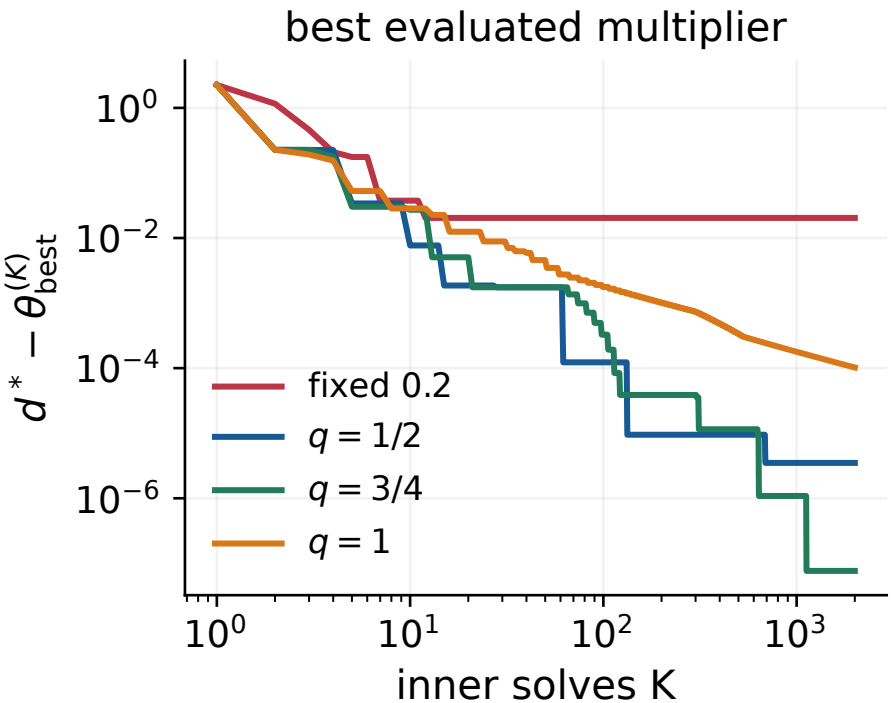
At $\mathbf{w}^* = (0, 1)$,

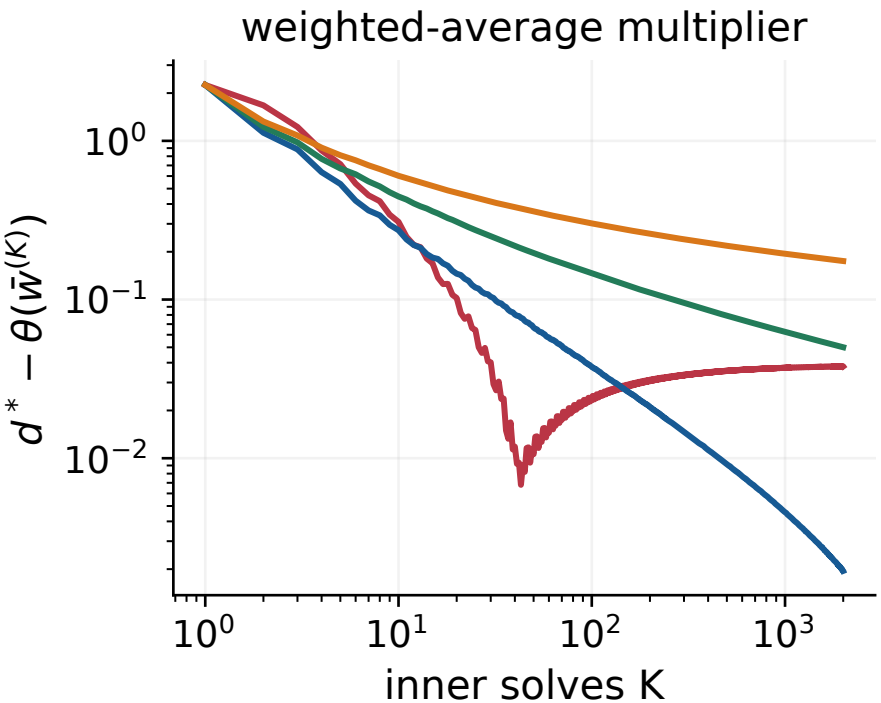
$$\partial^+\theta(\mathbf{w}^*) = [-\tfrac{5}{4}, \tfrac{3}{4}] \times \{0\}.$$

The selected residual is $(-5/4, 0)$, even though $\mathbf{0}$ also belongs to the set.

Comparing step rules on the same problem

Start $\mathbf{w}^{(0)} = (1, 0)$. Use $K = 2000$ exact inner solves. Compare fixed 0.2 with $\alpha^{(k)} = 0.5/(k+1)^q$.





At a finite budget, early errors can keep the weighted average less accurate.

Comparing step rules on the same problem (cont'd)

All entries are measured after the same 2000 inner solves.

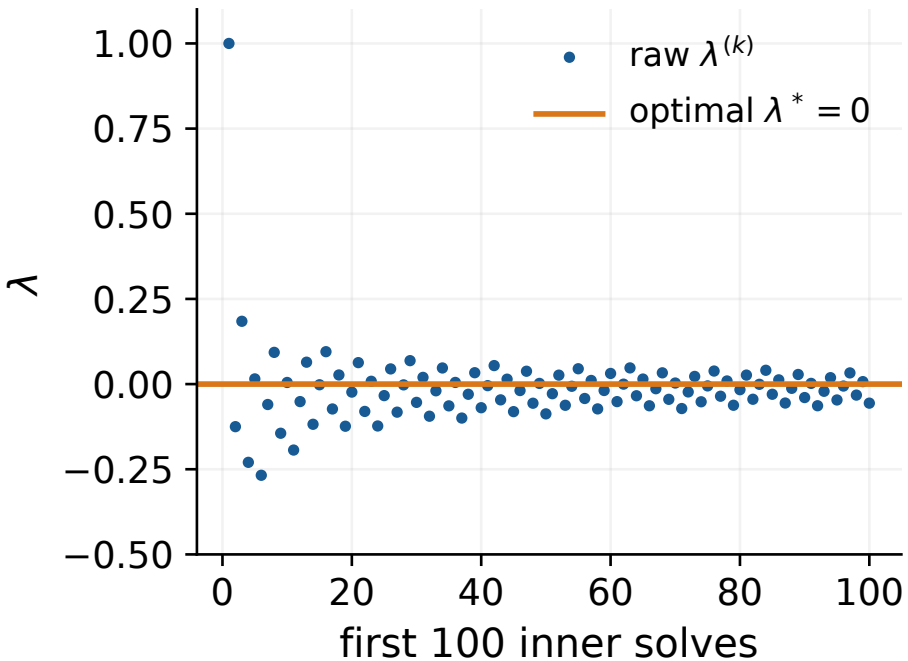
Step rule	Last $d^* - \theta$	Best $d^* - \theta$	Averaged $d^* - \theta$
Fixed 0.2	3.839×10^{-2}	2.038×10^{-2}	3.790×10^{-2}
$q = 1/2$	2.523×10^{-3}	3.498×10^{-6}	1.950×10^{-3}
$q = 3/4$	5.141×10^{-4}	7.656×10^{-8}	5.004×10^{-2}
$q = 1$	1.960×10^{-4}	1.023×10^{-4}	1.749×10^{-1}

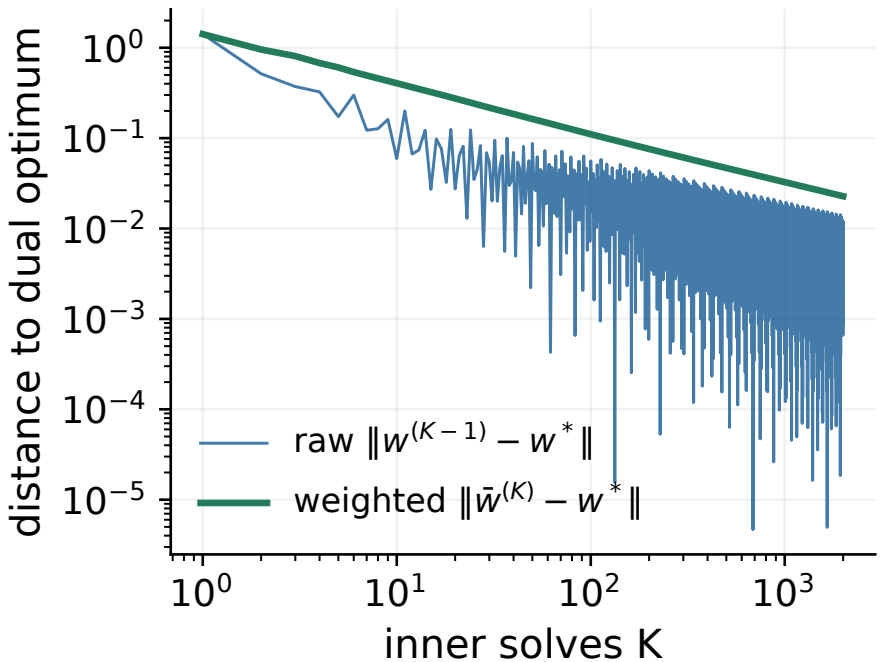
No universal ranking follows from this one test.

Dual iterates on the common test

For $q = 1/2$, use the same run:

$$\alpha^{(k)} = \frac{0.5}{\sqrt{k+1}}, \qquad \mathbf{w}^{(0)} = (1, 0), \qquad \mathbf{w}^{*} = (0, 1).$$

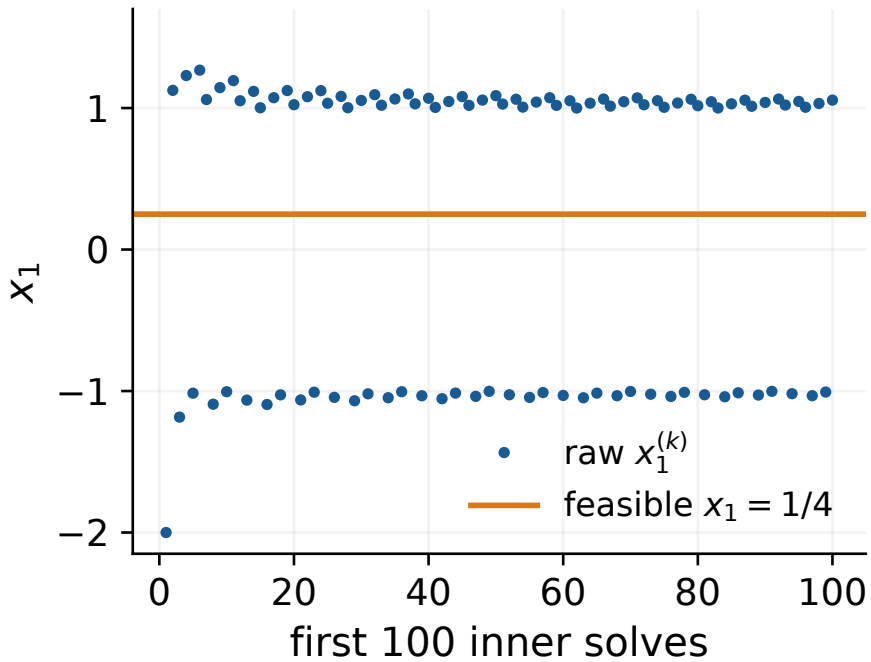


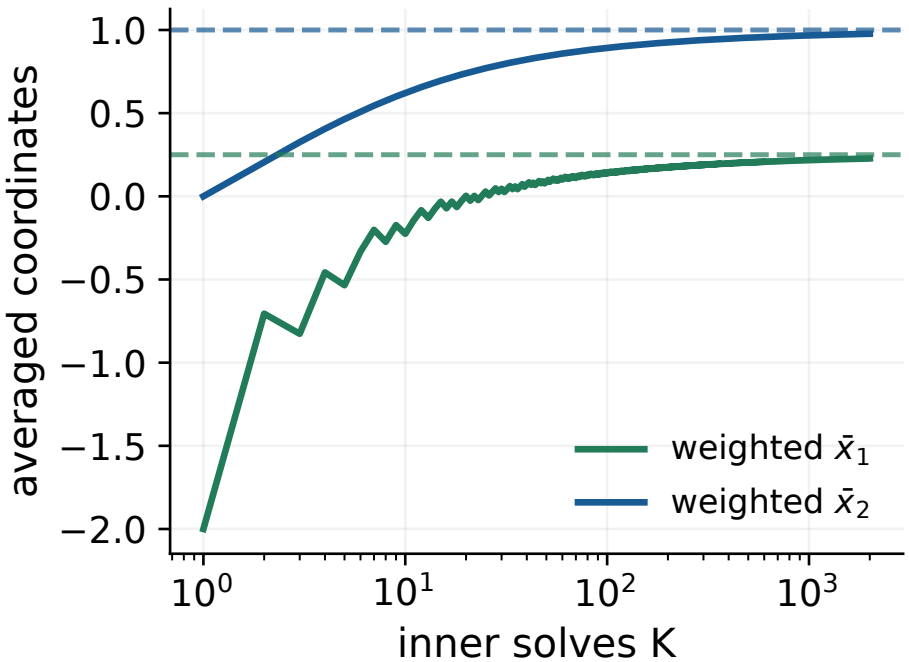


Raw multiplier oscillations shrink toward the dual optimum.

Dual success does not make each primal point feasible

For $q = 1/2$, form $\bar{\mathbf{x}}^{(K)} = S_K^{-1} \sum_{k<K} \alpha^{(k)} \mathbf{x}^{(k)}$.





Raw inner solutions keep switching across the feasible value.

Implementation: what can be certified?

- Stopping: use a budget or a genuine bound. If a primal feasible $\widehat{\mathbf{x}}$ is available,

$$f(\widehat{\mathbf{x}}) - \theta_{\text{best}}^{(K)} \leq \varepsilon_{\text{out}}$$

  certifies both objective errors. Small multiplier steps do not.
- Inexact inner solves: a certified global value error

$$0 \leq L(\widetilde{\mathbf{x}}^{(k)}, \mathbf{w}^{(k)}) - \theta(\mathbf{w}^{(k)}) \leq \varepsilon_{\text{in}}^{(k)}$$

  gives an approximate supergradient. A small stationarity residual alone is insufficient.
- Cost: a difficult global inner solve can dominate the method.

Comparison with previous constrained algorithms

Method	Inner task / model	Main implementation issue
QP	Quadratic-penalty minimization	Growing ρ
ALM	Augmented-Lagrangian minimization	Fixed, sufficiently large ρ
SQP	Quadratic program	Well-posedness; globalization
Barrier	Barrier subproblem	Strict interior; decreasing μ
Dual supergradient	Global Lagrangian minimization	Finite inner value; nonsmooth steps; primal recovery

NB. A positive duality gap still limits what any dual algorithm can recover.

Take-home messages

1. Geometry: a supergradient defines an upper support; it need not be an ascent direction.
2. Oracle: one global Lagrangian minimizer gives one residual supergradient.
3. Algorithm: project the multiplier step; least-norm selection is optional and can be expensive.
4. Convergence: nonsummable diminishing steps control best and averaged dual values; stronger assumptions give stronger conclusions.
5. Recovery: check primal feasibility and distinguish raw iterates from averages.